When Will Quantum Computers Beat Classical Simulation of a Real Quantum Magnet? A Quantum–Classical Crossover for the Neutron Spectrum of α-$RuCl_3$

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***Abstract*—Claims of quantum advantage for simulating materials are usually made for toy models at an arbitrary accuracy. We ask the question for a real material and an accuracy fixed by experiment: the dynamical structure factor $S(\mathbf{q},\omega)$ of the Kitaev magnet α-$RuCl_3$, resolved to the energy resolution of inelastic neutron scattering. We measure the cost of exact classical simulation on clusters of up to 18 spins and of matrix-product-state time evolution on cylinders, measure the Trotter error of the corresponding quantum circuits against the exact results, and feed the logical circuits to a surface-code resource estimator. Across three published exchange models, a fault-tolerant quantum computer overtakes exact state-vector simulation at $N^* \approx 36$–38 spins. For the Winter *et al.* model it needs 0.25 M (1.7 M) physical qubits at a physical error rate of 10^{-4} (10^{-3}), and at the crossover both machines need about a century, so the crossover marks where neither is practical rather than where quantum simulation becomes useful. Preparing the ground state from the best product state moves the crossover to $N^* \approx 92$–94. Even the qubit-minimal layout needs 29–34 thousand physical qubits at $p = 10^{-4}$, 29–34 times the 1000-qubit target of India's National Quantum Mission, and the two-qubit error reported for the 25-qubit QpiAI-Indus system is 4 times above the surface-code threshold. We claim no advantage today; every projection is stated with its assumptions, and code and data will be made public.**

***Index Terms*—quantum advantage, resource estimation, Kitaev materials, neutron scattering, Trotter error**

Plain-language summary. Neutron beams reveal how the magnetic moments inside a crystal move, and comparing those measurements with a model needs a computer simulation that becomes exponentially harder as the simulated piece of crystal grows. We estimate how large that piece must be before a future error-corrected quantum computer would finish the job faster than a classical computer, using the real accuracy of neutron experiments on the magnet α-$RuCl_3$. The answer is a few dozen magnetic atoms, but reaching it requires hundreds of thousands of physical qubits, far beyond any machine built or planned so far.

I. Introduction

Simulating quantum dynamics is the application most often named for quantum computers, yet most crossover studies cost a model Hamiltonian at an accuracy chosen by the authors. Yoshioka *et al.* [1] located the quantum–classical crossover for ground-state energies of two-dimensional model Hamiltonians, and Sun *et al.* [2] did so for the dynamics of the mixed-field Ising model, benchmarking against tensor-network and variational Monte Carlo methods. Two ingredients are still missing for a statement a materials scientist can use: a real material whose Hamiltonian is not exactly solvable, and an accuracy target set by an actual measurement rather than by a free parameter ε.

α-$RuCl_3$ is the leading candidate for Kitaev's honeycomb spin liquid [3], [4], [5]. Its inelastic neutron scattering (INS) spectrum shows a continuum whose origin is debated [6], [7], [8], [9], several exchange models compete [9], [10], [11], and exact diagonalization on small clusters is the standard tool for comparing them with data [11]. It is therefore a natural target for quantum simulation, and a natural test of when that simulation would pay off.

Contributions. (i) An accuracy target derived from the INS energy resolution: the time window, total simulated time and spectral tolerance all follow from one measured number (Sec. II). (ii) Measured classical cost of exact time-domain simulation of $S(\mathbf{q},\omega)$ on periodic clusters, including the degenerate ground manifolds that symmetric clusters carry, and of matrix-product-state time evolution, whose bond dimension we measure against exact results (Sec. III). (iii) Trotter errors measured on the experimental observable itself, for the exact circuit whose gates are counted, and a full surface-code resource estimate with ground-state preparation as a separate line item (Sec. IV). (iv) The crossover N^*, its sensitivity, and the gap between the hardware it needs and India's stated targets (Sec. V).

II. Model and accuracy target

We use the spin-1/2 J_1–K–Γ–Γ'–J_3 model on the honeycomb lattice [12], [9],

$$H = \sum_{\langle ij\rangle_\gamma} \Big[J_1\, \mathbf{S}_i\cdot\mathbf{S}_j + K S_i^\gamma S_j^\gamma + \Gamma\big(S_i^\alpha S_j^\beta + S_i^\beta S_j^\alpha\big) + \Gamma' \sum_{\mu=\alpha,\beta} \big(S_i^\gamma S_j^\mu + S_i^\mu S_j^\gamma\big)\Big] + J_3 \sum_{\langle\langle\langle ij\rangle\rangle\rangle} \mathbf{S}_i\cdot\mathbf{S}_j, \quad (1)$$

where $\gamma \in \{x,y,z\}$ labels the bond and α,β the other two axes. Our main set is the ab initio and neutron-fitted minimal model of Winter *et al.* [9], $(J_1, K, \Gamma, J_3) =$

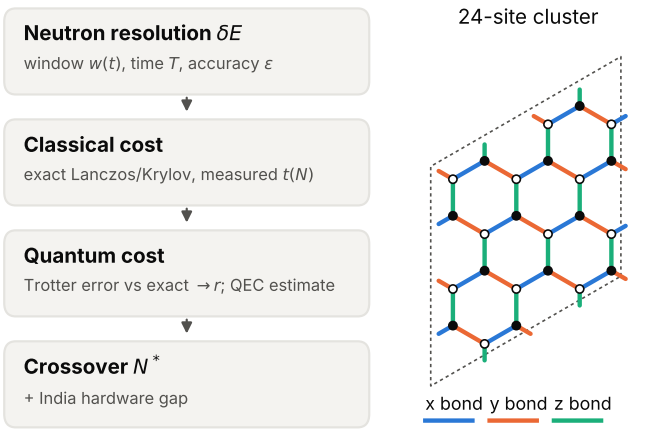


Fig. 1. The pipeline: the neutron resolution fixes the window, the total time and the accuracy; classical and quantum costs are measured or estimated for the same target and meet at N^*. Right: a 24-site periodic cluster with Kitaev bond labels (open and filled circles are the two sublattices; stubs leave the cell and re-enter on the opposite side).

$(-0.5, -5.0, 2.5, 0.5)$ meV (as tabulated in [13]); the K–Γ model of Ran *et al.*, $(K, \Gamma) = (-6.8, 9.5)$ meV [10], and the set Laurell and Okamoto fitted to the specific heat with exact diagonalization [11], $(J_1, K, \Gamma, \Gamma', J_3) = (-1.3, -15.1, 10.1, -0.12, 0.9)$ meV (as tabulated in [14]), test the sensitivity to the model; their energy scales are about two and three times larger.

Observable. We compute

$$S(\mathbf{q},\omega) = \frac{1}{2\pi}\int_{-T}^{T} dt\, w(t)\, e^{i\omega t}\, \frac{1}{N}\sum_{ij} e^{i\mathbf{q}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\langle \mathbf{S}_i(t)\cdot\mathbf{S}_j(0)\rangle \tag{2}$$

($\hbar = 1$, energies in meV). An instrument with Gaussian resolution of full width δE measures the true spectrum convolved with that Gaussian, which in the time domain is *exactly* the window $w(t) = e^{-t^2/2\tau^2}$ with $\tau = 2\sqrt{2\ln 2}/\delta E$. The correlation function is therefore needed only up to the time $T = \tau\sqrt{2\ln(1/c)}$ at which the window falls below a cutoff c (we use $c = 10^{-3}$), and sampling at $\Delta t = \pi/\omega_{\max}$ resolves energies up to $\omega_{\max} = 30$ meV. We take $\delta E = 0.50$ meV as a representative high-resolution value and report coarser resolutions in Table II; this gives $T = 17.5\,\text{meV}^{-1} = 11.5$ ps (the rough estimate $2\pi\hbar/\delta E$ gives 8.3 ps) and 169 time points. We demand that the resolution-convolved spectrum $\sum_a S^{aa}(\mathbf{q},\omega)$ be correct to $\varepsilon = 5\%$ of its peak value, a representative experimental uncertainty (Table II covers 1–10%), split equally between Trotter error and shot noise.

III. Classical cost

Clusters. For each N we enumerate all periodic supercells of $N/2$ unit cells by their Hermite normal forms and take the most compact one on which every nearest- and third-neighbour bond maps to a distinct pair of sites. Third-neighbour bonds are faithful only for $N \geq 14$, so we use $N = 14, 16, 18$ for the Winter set and $N = 14, 16$ for the others; larger clusters are left to future work.

Method. The model has no $U(1)$ symmetry, so we work in the full 2^N-dimensional space with a matrix-free Hamiltonian. The ground state is found by two-pass Lanczos; symmetric clusters have degenerate ground levels (6-fold for $N = 14$, matching three zigzag orientations times time reversal, and 2-fold for $N = 18$ with the Winter parameters), which we resolve by deflated Lanczos and average over, since a single Lanczos vector breaks translation symmetry. Each $S_j^b|0\rangle$ is propagated by short-iterative Lanczos [15] with an a-posteriori error bound of 10^{-9}, and overlaps with S_i^a give all correlators at each time; translation invariance reduces the reference sites j to one per sublattice. The code is checked against the exactly solvable Kitaev limit [3], [16] (conserved plaquettes, ultra-short-ranged static and dynamical correlations, ground-state energy within 2% of the thermodynamic value) and against the exact Lehmann sum for $N = 8$, which the time-domain result matches to 10^{-4} of the peak.

Measured cost. The full spectrum takes 7.1 min, 5.1 min and 49 min on a 4-core workstation for $N = 14, 16, 18$ (all ground states included). Per ground state, the cost follows $t \propto N^2 2^N$ (matrix–vector cost $\propto N2^N$ times a number of products $\propto \|H\|T \propto NT$); a free exponential fit gives a base of 2.13. Fig. 2 shows the computed spectrum for $N = 18$.

Matrix product states. We also evolve $S_j^z|0\rangle$ with two-site TDVP [17] in TeNPy [18], from a DMRG ground state, on cylinders of circumference $W = 2$ and 3 unit cells (16 and 18 spins). The MPO is built from the same bond list as the exact code, and with full bond dimension the two agree to 10^{-9}. For each bond dimension χ we find the longest window T' over which the error of the resolution-convolved spectrum, bounded by summing site-resolved errors against the exact result, stays below ε. For $W = 2$, $\chi = 16$, 32, 64 and 128 reach $T' = 1.2$, 3.1, 12.4 and 17.5 meV^{-1} (the full window); for $W = 3$, $\chi = 32$ and 64 reach only 1.1 and 2.5 meV^{-1}. Fitting $\ln\chi = c_W + s_W T'$ gives growth rates $s_W = 0.11$ and 0.51 meV, so the full window at $W = 3$ would already need $\chi \approx 1.4 \times 10^5$. We extrapolate the cost, measured per step as $\propto ND\chi^3$ (D the MPO bond dimension), to cylinders with $N = 2W^2$, with s_W and c_W linear in W and six evolutions (three components, two sublattices) per spectrum.

IV. Quantum cost

Circuit. One qubit per spin plus one ancilla. Each correlator is measured by the Hadamard test with controlled Paulis and an *uncontrolled* evolution between them [19], so the evolution circuit is the only non-Clifford part. The evolution is a second- or fourth-order Suzuki product formula [20] over the Pauli strings of H: five rotations $e^{-i\theta\sigma_i^a\sigma_j^b}$ per nearest-neighbour bond for the Winter set (nine if $\Gamma' \neq 0$) and three per third-neighbour bond, ordered by bond family, with neighbouring rotations about the same string fused.

Trotter error, measured. We simulate exactly this rotation sequence on a state vector, form the correlators as the Hadamard test would (evolving both $|0\rangle$ and $S_j^b|0\rangle$ with the circuit), and compute the error of the resolution-convolved spectrum against the exact result for $N = 14, 16, 18$, several

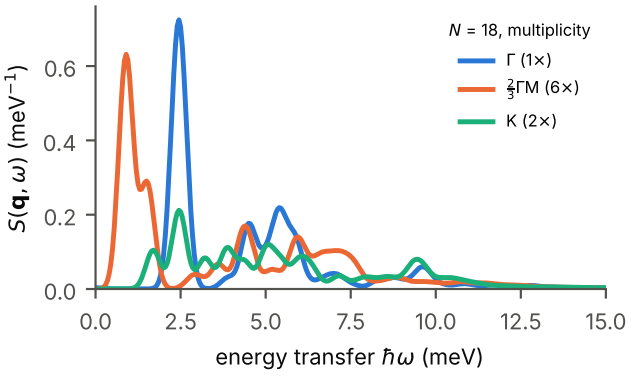


Fig. 2. Computed $\sum_a S^{aa}(\mathbf{q},\omega)$ at the symmetry-distinct momenta of the 18-site cluster (Winter parameters), convolved with the resolution $\delta E = 0.50\,\text{meV}$. Legend: momentum and number of equivalent momenta.

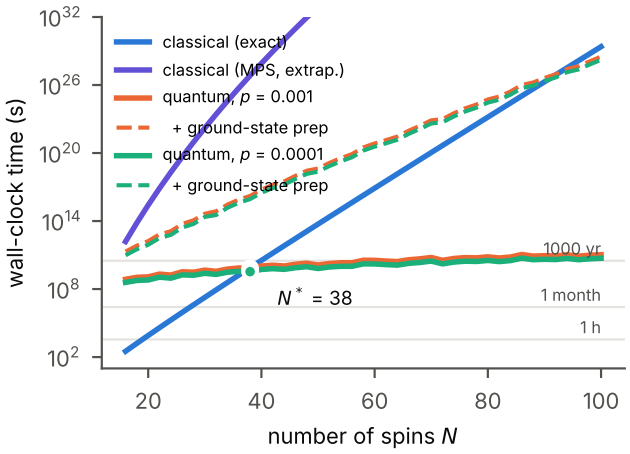


Fig. 3. Wall-clock time to obtain $S(\mathbf{q},\omega)$ at the experimental resolution and accuracy, against the number of spins. Classical: exact simulation, measured for $N \le 18$ and extrapolated as $N^2 2^N$, and MPS time evolution extrapolated from cylinders with $W = 2, 3$ (it leaves the plot near $N = 30$). Quantum: second-order Trotter circuits, surface code, without (solid) and with (dashed) ground-state preparation.

step sizes δt and three resolutions. The data follow $\varepsilon_{\text{Trot}} = A\,T\,\delta t^p$ with $p = 2.00$ for second order (30 points); fitting an explicit N^α gives $\alpha = -0.12$, so we take the largest prefactor among the measured sizes, $A = 0.73$, and no growth with N. This is the expected behaviour for an intensive, resolution-limited quantity; commutator bounds [21] would be far more pessimistic.

Shots. Each shot gives ± 1, so the variance of a correlator of spin-1/2 operators is at most $1/16n$ after n shots. Propagating this through the windowed transform gives $\text{Var}\,S \le 3N \sum_k (w_k \Delta t)^2/(32\pi^2 n)$, and we choose n so that two standard deviations stay below half of $\varepsilon S_{\max}$. Here $S_{\max}$ is the peak of the inelastic spectrum ($\omega > 2\delta E$), excluding the quasi-elastic weight that piles up at the zigzag ordering vector on clusters that fit that order, and taking the smallest peak over the measured sizes ($S_{\max} = 0.23\,\text{meV}^{-1}$ for the Winter set, $n \approx 1.9 \times 10^4$). With one reference site per sublattice, three spin components and real and imaginary parts, $12N$ circuits are needed per time point, 7.7×10^4 in total at $N = 38$. All are independent, so they could run on many machines at once; we compare one quantum computer with one classical workstation.

Error correction. Logical counts (qubits, rotations, rotation depth from as-soon-as-possible scheduling) are passed to Microsoft's resource estimator [22] with a surface code [23], gate-based qubits with nanosecond gate times and physical error rates $p = 10^{-3}$ and 10^{-4}, and a total error budget of 1% per circuit; rotation synthesis [24] and magic-state factories are included by the estimator. One machine runs all circuits sequentially.

Ground-state preparation. We use a simple model of phase-estimation projection from a product state with overlap γ: an evolution time $2\pi \ln(1/\delta)/\Delta$ per attempt with controlled evolution at twice the cost, and $1/\gamma^2$ attempts per shot; near-optimal methods reduce the latter to $1/\gamma$ [25]. From exact results we take the mean gap $\Delta = 0.37\,\text{meV}$ and the best product-state overlap, optimised over all single-spin directions, which decays as $\gamma \propto e^{-0.21N}$ over $N = 14, 16, 18$. This line item is reported separately because it is the least certain.

TABLE I

FAULT-TOLERANT RESOURCES AT THE CROSSOVER FOR THE THREE PARAMETER SETS (SECOND-ORDER TROTTER, $\delta E = 0.50\,\text{MEV}$, $\varepsilon = 5\%$). r: TROTTER STEPS FOR THE LONGEST TIME; d: CODE DISTANCE; RUNTIME EXCLUDES GROUND-STATE PREPARATION.

Set	p	N^*	r (steps)	T states per shot	d	Phys. qubits	Runtime at N^*
Winter	10^{-3}	38	396	7.2×10^6	19	1.7 M	209 yr
Winter	10^{-4}	38	396	7.2×10^6	9	0.25 M	109 yr
Ran	10^{-3}	36	1757	1.1×10^7	19	5.2 M	55 yr
Ran	10^{-4}	36	1757	1.1×10^7	9	0.41 M	28 yr
Laurell	10^{-3}	38	2751	7.9×10^7	21	3.5 M	558 yr
Laurell	10^{-4}	38	2751	7.9×10^7	11	0.60 M	270 yr

V. CROSSOVER AND INDIA ROADMAP

Fig. 3 shows the result for the Winter set. The quantum cost grows only polynomially, but its prefactor is large: at N^* a single circuit runs for seconds, and $\sim 10^5$ circuits each need $\sim 10^4$ shots. The curves cross at $N^* = 38$ for both error rates, where the quantum computer needs 109 yr ($p = 10^{-4}$) or 209 yr ($p = 10^{-3}$) against 241 yr classically (Table I). The crossover is therefore a statement about scaling, not about a useful computation: practical simulation at N^* needs both far more parallel hardware and better algorithms on each side. Fourth-order formulas give the same N^*, because fewer steps are offset by five times as many rotations per step. Including ground-state preparation moves the crossover to $N^* = 92$–94, because the product-state overlap shrinks exponentially; better initial states [25] are the main lever. The extrapolated MPS cost lies above the exact cost at every size we consider (Fig. 3), so the quantum computer beats it from the smallest size on, but it is never the cheaper classical method: with the long time

TABLE II

SENSITIVITY OF THE CROSSOVER: ONE ASSUMPTION CHANGED AT A TIME FROM THE BASELINE (WINTER SET, $\delta E = 0.50$ MEV, $\varepsilon = 5\%$, SECOND-ORDER TROTTER, NANOSECOND GATES WITH $p = 10^{-4}$, ONE CLASSICAL WORKSTATION). RUNTIME AND PHYSICAL QUBITS ARE THOSE OF THE QUANTUM COMPUTER AT N^*, WITHOUT GROUND-STATE PREPARATION.

Variation	N^*	Runtime	Phys. qubits
Baseline (§V)	38	109 yr	0.25 M
Accuracy $\varepsilon = 1\%$	44	9927 yr	0.24 M
Accuracy $\varepsilon = 2\%$	42	2005 yr	0.19 M
Accuracy $\varepsilon = 10\%$	36	20 yr	0.2 M
Resolution $\delta E = 1$ meV	36	25 yr	0.2 M
Resolution $\delta E = 2$ meV	34	2 yr	0.12 M
Physical error $p = 10^{-3}$	38	209 yr	1.7 M
μs gates, $p = 10^{-4}$	50	308374 yr	0.31 M
μs gates, $p = 10^{-3}$	50	628735 yr	1.1 M
Fourth-order Trotter	38	89 yr	0.25 M
With ground-state prep.	92	–	–
Classical $10^2\times$ faster	44	173 yr	0.24 M
Classical $10^4\times$ faster	52	278 yr	0.26 M
Ran parameters	36	28 yr	0.41 M
Laurell–Okamoto parameters	38	270 yr	0.6 M

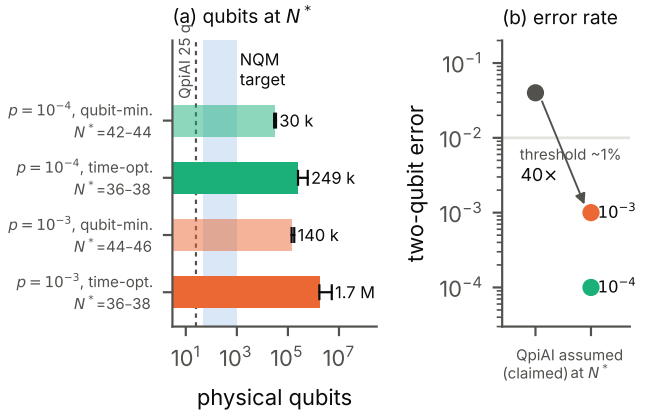


Fig. 4. (a) Physical qubits needed at the crossover for the time-optimal and the qubit-minimal surface-code layout at two physical error rates, against the National Quantum Mission target of 50–1000 physical qubits [26] and the 25-qubit QpiAI-Indus system [27]. Bars: Winter set; whiskers: range over the three parameter sets. (b) The assumed physical error rates against the reported two-qubit error of QpiAI-Indus.

window that the neutron resolution demands, entanglement growth rather than system size limits MPS, and the crossover is set by exact simulation.

Sensitivity. Table II varies one assumption at a time; across the full grid of 192 combinations (three parameter sets, three resolutions, four accuracies, four qubit models and both Trotter orders where measured) N^* stays between 32 and 56 against one workstation. *Experiment:* a tighter accuracy or a finer resolution moves N^* by a few spins but changes the quantum run time at N^* by orders of magnitude, from years at $\delta E = 2$ meV to millennia at $\varepsilon = 1\%$, because the shot count grows as ε^{-2} and the circuits lengthen with T. *Quantum hardware:* the physical error rate sets the qubit count but hardly moves N^*; slow (microsecond) gates, as in trapped ions, make every shot about 1500 times longer and push N^* to the largest values in the grid. *Classical hardware:* each factor of two in classical speed costs roughly one spin, so a large cluster 10^4 times faster than our workstation moves N^* by about 14. *Model:* the Ran *et al.* and Laurell–Okamoto sets (exact data for $N = 14, 16$) have Trotter prefactors $A = 16$ and 24, twenty to thirty times that of the Winter set, yet cross over at nearly the same N^*: the step count grows only as $A^{1/2}$, and the circuit and shot counts, which dominate, barely change. With only two sizes the overlap trend of these sets is not determined (for Laurell–Okamoto it even rises from $N = 14$ to 16), so we quote the crossover with ground-state preparation for the Winter set only.

India roadmap. The National Quantum Mission aims at intermediate-scale machines with 50–1000 physical qubits within eight years [26]; the superconducting QpiAI-Indus system has 25 qubits with a reported two-qubit fidelity of 96% [27]. We map the hardware needed at the crossover for all three parameter sets and two surface-code layouts (Fig. 4): the time-optimal layout of Table I, with as many magic-state factories as shorten the run, and a qubit-minimal layout with one factory, which runs 60–75 times longer per shot and therefore crosses over later. *Qubits.* At $p = 10^{-4}$ the crossover needs 249–597 thousand physical qubits time-optimally ($N^* = 36$–38) and 29–34 thousand in the qubit-minimal layout ($N^* = 42$–44, but 5512–38363 years of run time); at $p = 10^{-3}$ these become 1708–5218 and 140–177 thousand. Against the mission's 1000-qubit target the smallest of these machines is 29–34 times larger, about 5 doublings of qubit count beyond 2031; the time-optimal one is 8–9 doublings away. *Error rate.* A 96% fidelity is a 4% error, 4 times above the ∼1% surface-code threshold, so no number of such qubits can run this computation; reaching $p = 10^{-3}$ or 10^{-4} needs a further 40 or 400 times lower error. The error rate is the binding constraint and comes first: below threshold every factor of ten in p cuts the qubit count by a factor of five to thirteen in our estimates, which is worth more than any of the doublings above. For India's roadmap this application therefore asks first for two-qubit errors near 10^{-4}, and only then for qubit counts of order 10^4–10^5, well beyond the 2031 target.

Hardware check. To see where present hardware stands, we run the same Trotter circuit on 12 qubits. The Hadamard-test protocol, with ground-state preparation, is far too deep, so we use the standard quench: the Néel state of an open 12-spin honeycomb flake with the Winter couplings is evolved and every spin is measured in the z basis. The Qiskit [28] circuit reproduces our state-vector Trotter evolution exactly. Transpiled to IBM's Heron processor `ibm_torino` it needs 180 two-qubit gates per Trotter step. On an emulator with that processor's published noise calibration (not yet a device run), the measured magnetizations deviate from the exact ones by up to 0.39 after a single step at $t = 0.125\,\text{meV}^{-1}$, only 0.7%

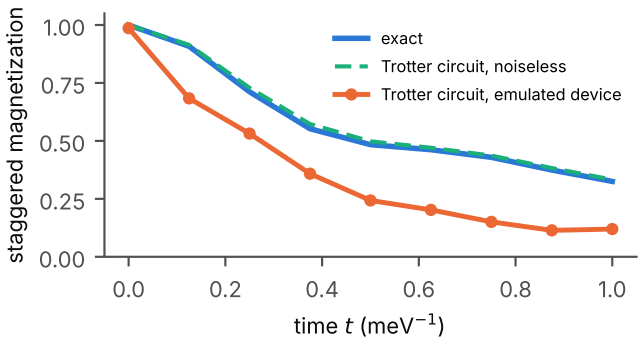


Fig. 5. Hardware check on 12 qubits: staggered magnetization after a Néel-state quench on a 12-spin flake (Winter couplings), from exact evolution, from the noiseless Trotter circuit, and from the same circuit transpiled to `ibm_torino` and run on an emulator with that processor's published noise calibration (2000 shots per point; one Trotter step per $0.125\,\mathrm{meV}^{-1}$, 180–1353 two-qubit gates).

of the window T, whereas the noiseless circuit stays within 0.04; after 1353 two-qubit gates only 37% of the staggered magnetization survives (Fig. 5). Without error correction the experiment-defined target is out of reach by orders of magnitude in circuit depth, which is why the crossover assumes a fault-tolerant machine.

VI. Limitations and Outlook

Classical baseline. Exact simulation is not the strongest conceivable classical method, so our N^* is a lower bound on where advantage could begin. Translation symmetry would save a further factor $\sim N/2$, and continued-fraction Lanczos [29] avoids time stepping altogether for a fixed broadening. Our MPS estimate rests on two short cylinders, a bound rather than the momentum-resolved error, and a linear fit of $\ln \chi$ in time and width, so it is crude; PEPS, time-dependent variational Monte Carlo and neural quantum states were not run. Each could push N^* up, although long-time dynamics in two dimensions is where they struggle most. *Extrapolation.* Classical costs and Trotter errors are extrapolated from $N \leq 18$ (Winter) or $N \leq 16$ (the other sets); the absence of an N trend in the Trotter error rests on few sizes with strong finite-size effects, and so does the reference peak $S_{\max}$, which varies by a factor of three between sizes. *Measurement.* Measuring $S(\mathbf{q}, t)$ directly rather than site-resolved correlators, or sharing shots between circuits, could cut the circuit count by up to a factor N. *Physics.* We assume zero temperature, ignore the neutron polarization factor and use parameter sets that are themselves uncertain. *Ground-state preparation* is modelled coarsely and dominates when included. Despite these caveats, the experiment-defined target turns "quantum advantage for materials" into a concrete, checkable number, and our open code lets each assumption be replaced.

Code and data will be released publicly upon acceptance.

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